

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 01

Combinational Circuit



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Recap of Previous Lecture

Question Practice



Topics to be Covered

Comb. Ckt.

H.A, F.A.

[Type of Digital Circuit]



- Combinational Circuit
- Sequential Circuit

Comb. CKT

- O/P depends on present i/p values only.
- There is no memory.
- There is no feedback.

eg. H-A, F-A, H-S, F-S, MUX, Decoder, Encoder etc.

Seq. CKT.

- O/P depends on present as well as past i/p values.
- There is memory.
- There is feedback.

eg. Counter, Registers, Johnson Counter, etc.

[How to Design Combinational Circuit]



- *** firstly identify the no. of i/p lines & o/p lines.
- Write down the truth table b/w i/p lines & o/p lines.
- Now write down o/p_s using SOP or POS in terms of i/p_s.
- Simplify o/p boolean functions using theorems or K-Map.
- Implement simplified o/p boolean functions using logic gates.



Design a comb. ckt which takes 3-i/p_s and generate o/p '1' if i/p is either divided '2' or '3'.

→ i/p lines → 3-i/p_s → x, y, z

o/p line → 1-o/p → f

x	y	z	f
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

$$f(x, y, z) = \sum(0, 2, 3, 4, 6,)$$



Q. Design a Comb. CKt which takes 3-bit i/p and generate square of i/p number

→ i/p lines → 3-i/p lines → x, y, z

O/p lines → 6-o/p lines $y_5, y_4, y_3, y_2, y_1, y_0$

$$y_5[x, y, z] = \Sigma(6, 7)$$

$$= xy\bar{z} + xyz$$

$$= xy$$

$$y_4[x, y, z] = \Sigma(4, 5, 7)$$

x	y	z	y_5	y_4	y_3	y_2	y_1	y_0
0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	1
0	1	0	0	0	0	1	0	0
0	1	1	0	0	1	0	0	1
1	0	0	0	1	0	0	0	0
1	0	1	0	1	1	0	0	1
1	1	0	1	0	0	1	0	0
1	1	1	1	1	0	0	0	1



→ decimal

$$\begin{array}{r} 786 \\ 147 \\ + \quad 11 \\ \hline 0933 \end{array}$$

→ binary no. system:

$$\begin{array}{r} 1101 \rightarrow \\ + 0101 \rightarrow \\ \hline 10010 \end{array}$$

Octal no. $\xrightarrow{8^2 \ 8^1 \ 8^0}$ base = 8, digits (0-7)

$$\begin{array}{r} 145 \\ + 243 \\ \hline 0410 \end{array} \rightarrow (64 + 32 + 5)_{10} = (101)_{10}$$
$$\rightarrow (128 + 32 + 3)_{10} = (163)_{10}$$
$$\rightarrow (256 + 8 + 0)_{10} = (264)_{10}$$

$$\begin{array}{cccc} a_3 & a_2 & a_1 & a_0 \\ b_3 & b_2 & b_1 & b_0 \\ \hline c_3 & c_2 & c_1 & c_0 \\ s_3 & s_2 & s_1 & s_0 \end{array}$$

[Standard Combinational Circuits]

- Half Adder
- Half Subtractor
- Full Adder
- Full Subtractor

[Half Adder]



- What is half adder and what are input lines and what are output lines?

→ i/p lines → 2-i/p lines → x, y → corresponding bits of the two no.s being added

$$\begin{array}{l} x \rightarrow a_0 \\ y \rightarrow b_0 \end{array} \quad \text{or} \quad \begin{array}{l} x \rightarrow a_1 \\ y \rightarrow b_1 \end{array} \quad \text{or} \quad \begin{array}{l} x \rightarrow a_2 \\ y \rightarrow b_2 \end{array}$$

→ o/p lines → 2-o/p lines

→ Sum o/p lines S

→ Carry o/p line C → Carry o/p will be '1' if addition ($x+y$) is equal to base or greater than base 2.

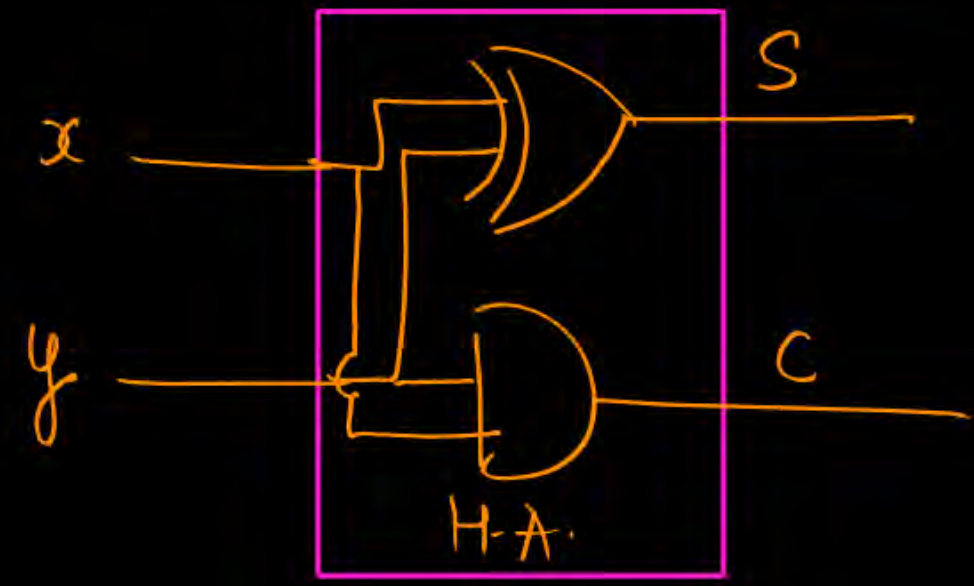
Input lines		Output lines	
x	y	S-o/p	C-o/p
0	0	0	0
1	0	1	0
2	1	1	0
3	1	0	1

$$S(x, y) = \sum(1, 2) = \overline{x}y + x\overline{y} = x \oplus y$$

→ Non self dual

$$C(x, y) = \sum(3) = x \cdot y$$

→ Non-self dual

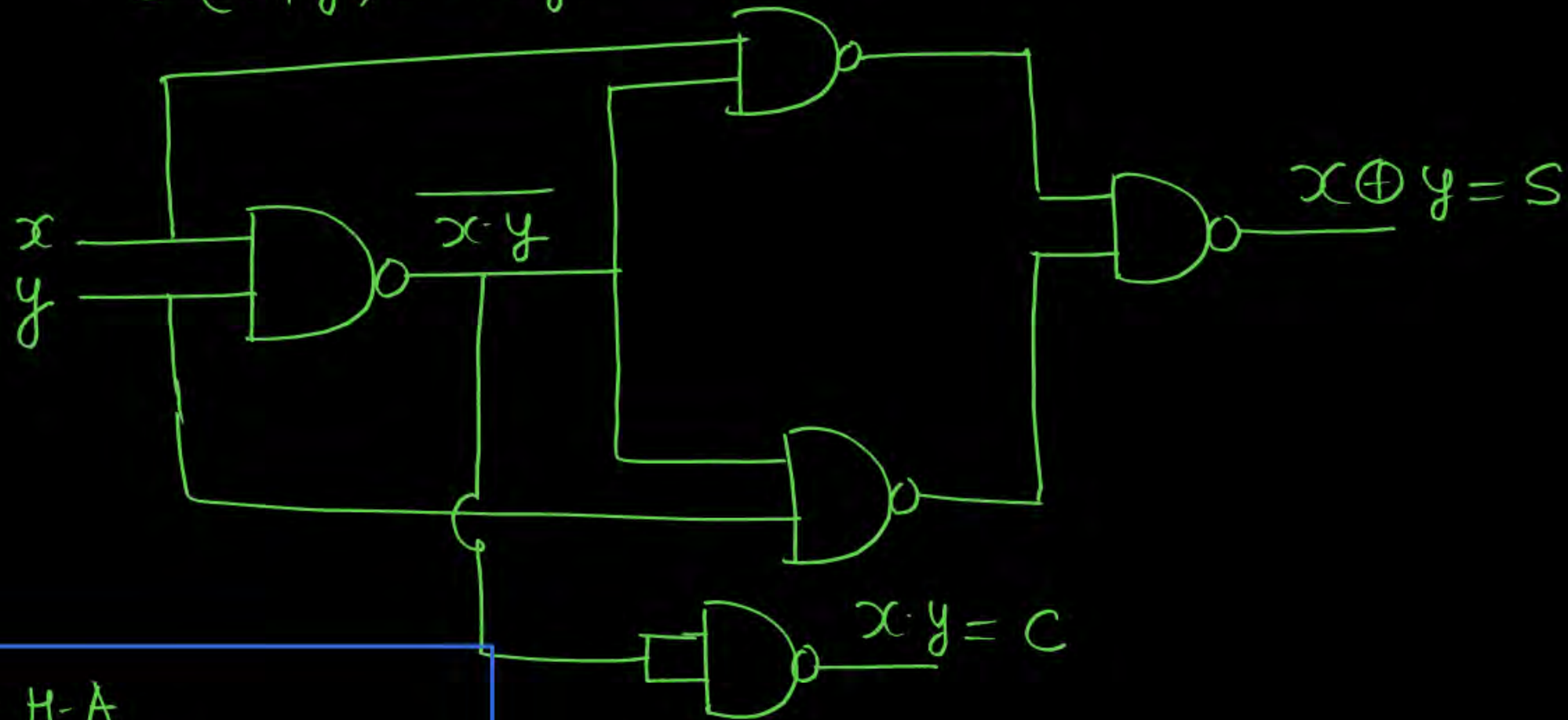




- Using NAND gates :

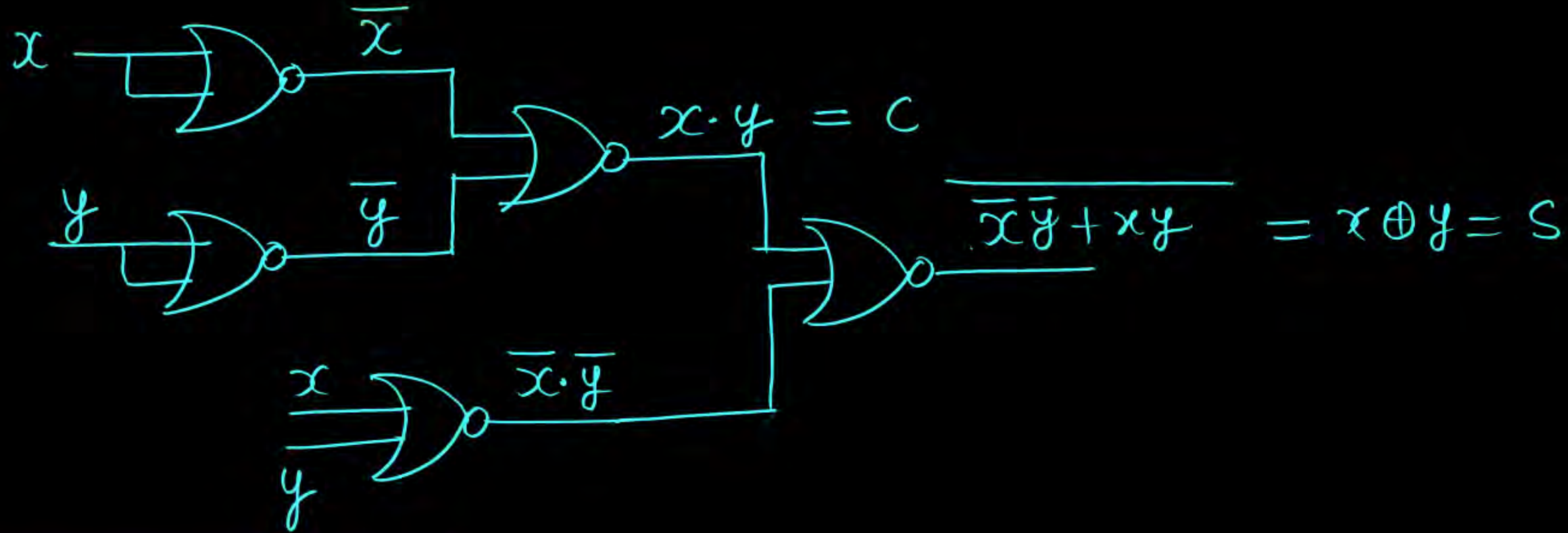
$$x, y \longrightarrow S(x, y) = x \oplus y$$

$$C(x, y) = x \cdot y$$



• To implement H-A
5 (2-IP NAND gates) are required

- Using NOR gates :



- To implement H. A.
we require 5 [2-1/P NOR]
gates.

[Full Adder]



- What is full adder and what are input lines and what are output lines?

→ i/p lines → 3-i/p lines → x, y → corresponding bits of the two no. being added
 z → Carry o/p generated from previous position
 $x, y \rightarrow a_2, b_2$ then $z \rightarrow C_2$ (Carry o/p from a_1, b_1 position)
 $x, y \rightarrow a_3, b_3$, then $z \rightarrow C_3$ (Carry o/p from a_2, b_2 position).

→ o/p lines → 2-o/p lines
├→ Sum o/p S
└→ Carry o/p C → Carry o/p will be '1' if $(x+y+z)$ is greater than or equal to base 2, otherwise '0'.



Input lines			Output lines	
x	y	z	S-o/p	C-o/p
0	0	0	0	0
1	0	1	1	0
2	0	0	1	0
3	0	1	0	1
4	1	0	1	0
5	1	0	0	1
6	1	1	0	1
7	1	1	1	1

$$S(x, y, z) = \sum(1, 2, 4, 7) = \pi(0, 3, 5, 6)$$
$$= x \oplus y \oplus z$$

↓
Self dual

$$C(x, y, z) = \sum(3, 5, 6, 7) = \pi(0, 1, 2, 4)$$

↓
Self dual

→ Majority I/P CKT

Thank you

GW
Soldiers !

